WORKED SOLUTION

STEP III Specimen paper, Question 6

Pure • Integration by parts

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Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/spec-s3-q6

The key idea

The integrand $\operatorname{sech}^{n+2}u \sinh^2 u$ splits naturally as $\sinh u$ times $\sinh u \operatorname{sech}^{n+2}u$, and the second factor is, up to a constant, the derivative of $\operatorname{sech}^{n+1}u$. That makes integration by parts the obvious move. For the reduction formula, the identity $\cosh^2 u - \sinh^2 u = 1$ lets us rewrite the same integral as a difference of two of the I's.

Solution

The integral identity

First, a derivative we shall need. Since $\operatorname{sech} u = (\cosh u)^{-1}$, the chain rule gives

$$\frac{\mathrm{d}}{\mathrm{d}u}\operatorname{sech}^{n+1}u = -(n+1)\cosh^{-n-2}u \sinh u = -(n+1)\operatorname{sech}^{n+2}u \sinh u.$$

So we integrate by parts with

$$f = \sinh u, \qquad g' = \operatorname{sech}^{n+2}u \sinh u, \qquad \text{so} \qquad f' = \cosh u, \qquad g = -\frac{\operatorname{sech}^{n+1}u}{n+1}.$$

This gives

$$\begin{aligned}\int_0^\infty \operatorname{sech}^{n+2}u \sinh^2 u \,\mathrm{d}u &= \left[-\frac{\sinh u \operatorname{sech}^{n+1}u}{n+1}\right]_0^\infty + \frac{1}{n+1}\int_0^\infty \cosh u \operatorname{sech}^{n+1}u \,\mathrm{d}u \\ &= -\frac{1}{n+1}\left[\tanh u \operatorname{sech}^n u\right]_0^\infty + \frac{1}{n+1}\int_0^\infty \operatorname{sech}^n u \,\mathrm{d}u,\end{aligned}$$

using $\sinh u \operatorname{sech}^{n+1}u = \tanh u \operatorname{sech}^n u$ and $\cosh u \operatorname{sech}^{n+1}u = \operatorname{sech}^n u$.

? **Ask yourself** Why does the boundary term vanish, and where is $n > 0$ used?

At $u = 0$ we have $\tanh 0 = 0$. As $u \to \infty$, $\tanh u \to 1$ while $\operatorname{sech} u = 2/(\mathrm{e}^u + \mathrm{e}^{-u}) \to 0$, so $\operatorname{sech}^n u \to 0$ *because* $n > 0$. (For $n = 0$ the term would tend to 1, not 0.) The same

fact, $\operatorname{sech}^n u < 2^n e^{-nu}$, shows that I_n converges for $n > 0$. Hence the boundary term is zero and

$$\int_0^\infty \operatorname{sech}^{n+2} u \sinh^2 u \, du = \frac{1}{n+1} I_n,$$

as required.

The reduction formula

Now evaluate the same integral a second way. Since $\sinh^2 u = \cosh^2 u - 1$,

$$\operatorname{sech}^{n+2} u \sinh^2 u = \operatorname{sech}^{n+2} u \cosh^2 u - \operatorname{sech}^{n+2} u = \operatorname{sech}^n u - \operatorname{sech}^{n+2} u.$$

Both pieces have convergent integrals over $[0, \infty)$ (as $n > 0$), so

$$\int_0^\infty \operatorname{sech}^{n+2} u \sinh^2 u \, du = I_n - I_{n+2}.$$

Comparing with the first result,

$$I_n - I_{n+2} = \frac{1}{n+1} I_n \quad \Longrightarrow \quad I_{n+2} = \left(1 - \frac{1}{n+1}\right) I_n = \frac{n}{n+1} I_n,$$

that is, $(n+1) I_{n+2} = n I_n$, as required.

The value of I_6

The reduction formula steps n by 2, so we need a starting value with n even. The easiest is

$$I_2 = \int_0^\infty \operatorname{sech}^2 u \, du = \Big[\tanh u\Big]_0^\infty = 1 - 0 = 1.$$

(We can't start from I_0: it diverges, and the formula with $n = 0$ would give the false statement $I_2 = 0$. This is exactly where $n > 0$ matters.) Then

$$I_4 = \frac{2}{3} I_2 = \frac{2}{3},$$
$$I_6 = \frac{4}{5} I_4 = \frac{4}{5} \cdot \frac{2}{3} = \frac{8}{15}.$$

So

$$I_6 = \frac{8}{15}.$$



Checks

- The sequence $I_2, I_4, I_6 = 1, \frac{2}{3}, \frac{8}{15}$ is decreasing, as it must be: $0 < \operatorname{sech} u \leq 1$, so higher powers are smaller.
- Starting instead from $I_1 = \int_0^\infty \operatorname{sech} u \, \mathrm{d}u = \left[2 \arctan(\mathrm{e}^u)\right]_0^\infty = \frac{\pi}{2}$, the formula gives $I_3 = \frac{1}{2} I_1 = \frac{\pi}{4}$, which agrees with direct numerical integration.
- Simpson's rule on $[0, 60]$ gives $I_6 \approx 0.5333333$, matching $\frac{8}{15}$; both the identity and the reduction formula also check numerically for non-integer n, such as $n = 0.5$ and $n = 2.5$.

Common slips

- Getting the sign or the factor $n + 1$ wrong in $\frac{\mathrm{d}}{\mathrm{d}u} \operatorname{sech}^{n+1} u$: $\operatorname{sech} u$ is decreasing for $u > 0$, so the derivative must be negative.
- Not justifying why the boundary term vanishes at infinity. This is the one place the condition $n > 0$ is needed.
- Writing the answer as $\frac{4}{5} \cdot \frac{2}{3} \cdot I_2$ without evaluating I_2.

Beyond STEP

Reduction formulae like this one are the same structure as Wallis's formula: with $t = \tanh u$ (or $\sin \theta = \tanh u$, the Gudermannian substitution), I_n becomes $\int_0^{\pi/2} \cos^{n-1} \theta \, \mathrm{d}\theta$, and the recurrence is Wallis's. At university these integrals reappear as values of the Beta function, $I_n = \frac{1}{2} B\left(\frac{n}{2}, \frac{1}{2}\right)$, which extends the recurrence to all real $n > 0$, and the care we took over the boundary term is the beginning of the theory of improper integrals. For a rigorous treatment, see Abbott, *Understanding Analysis*; for many more problems in this style, Siklos, *Advanced Problems in Mathematics*.

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